Project Timeline:

- **M0**: Team Preference Survey — Due Mar 06 (Week 9 – Friday)
- **M1**: Project Proposal — Due Mar 13 (Week 10 – Friday)
- **M2**: Progress Presentation — Due April 03 (Week 13 – Friday)
- **M3**: FINAL Deliverable — Due April 15 (Week 15 – Wednesday)

Your Final Project is worth **30% of your total course grade**, distributed as follows:

- **Milestone M1: Project Proposal – 2%**
- **Milestone M2: Progress Presentation – 5%**
- **Milestone M3: Final Report & Code – 23%**

PROJECT Details

This is a team-based Applied Research project where you will collaborate closely with your group (teams of 3) to develop a high-impact, portfolio-worthy Generative AI project.

What is Applied Research? Your project should sit at the intersection of research and engineering. You should build something that works, but also ask a meaningful question and answer it with rigor. Strong projects don't just produce a working system — they explain *why* it works, *how well* it compares to alternatives, and *what breaks* when conditions change.

We expect significant effort and commitment, but rather than seeking a perfect project, we are interested in how effectively you explore different generative techniques, identify challenges, and develop strategies to improve your results.

Your final submission should demonstrate:

- A clearly defined problem and research question relevant to Generative AI.
- Multiple approaches explored with comparative analysis (e.g., prompting vs. fine-tuning, different retrieval strategies, various model architectures).
- Honest discussion of what worked, what didn't, and why — including challenges faced and iterative improvements made throughout the project.

Scope: Given three team members working over approximately 5-6 weeks, we expect a scope equivalent to at least a short workshop-style paper — if not a full conference submission, but a complete applied research cycle from problem definition through evaluation. The goal is to give you hands-on experience with the potential for publication or real-world impact. We try our best to support those actually interested in submitting to a conference.

While we expect equal contribution from all team members, individual grades may vary based on peer assessments and effective teamwork (see below).

Effective Teamwork & Peer Assessment

Effective teamwork is essential to this project. To set your team up for success, we require two things: a **Team Contract** at the start and **Peer Assessments** at the end.

Team Contract

At the beginning of your project, each team will complete a Team Contract using the provided template. This contract establishes baseline expectations for your team, including communication norms, meeting schedules, task distribution, and how you will handle disagreements. Think of it as your team's operating agreement — it protects everyone and prevents misunderstandings later. All team members must sign the contract and submit it alongside your M1 Proposal.

Peer Assessment

At the end of the term, each team member will individually evaluate the contributions of their teammates and self-assess their own efforts. **M3 (Final Assessment):** A full peer assessment submitted alongside your final deliverable. This assessment will directly factor into your individual project grade.

How it affects your grade: If peer assessments indicate significantly unequal contribution, individual grades may be adjusted up or down by up to 20% of the project component grade.

If your self-assessment significantly diverges from how your teammates evaluate you, this discrepancy will be noted and may (positively) affect your individual grade adjustment. This rewards honest self-awareness.

Your peer feedback should be honest, constructive, and fair. Evaluate each teammate on the following criteria:

- **Preparation** – Were they prepared for meetings and discussions?
- **Contribution** – Did they actively contribute to discussions and project work?
- **Collaboration** – Did they encourage and respect others' ideas?
- **Adaptability** – Were they flexible and cooperative during disagreements?

Confidentiality: All peer assessments are confidential and will only be reviewed by the instructor and TAs. Your teammates and other students will not see your individual responses.

M1: Project Proposal

For your first milestone, you will submit a **Final Project Proposal** (**1-2 pages, 500-1000 words, max 2 pages**). This proposal should outline the foundation of your Generative AI project and include the following components:

1. Introduction & Motivation
Provide a brief introduction to your chosen topic, its relevance to Generative AI, and why your team chose this problem. Keep this concise — a few sentences are sufficient.

2. Research Question / Problem Statement Clearly state the question your project aims to answer or the problem it aims to solve. A strong research question goes beyond "we will build X" — it should articulate what you are investigating, comparing, or evaluating. For example: *"How does RAG-based retrieval compare to fine-tuning for domain-specific medical question answering?"*

3. Scope of the Project
Define the **scope and focus** of your project. Given the **limited timeframe**, specify:

- What aspects of the problem you **will** explore.
- What aspects you **will not** address.

4. Plan of Action
Outline a **high-level roadmap** for the next four weeks, including:

- Key **tasks** and **milestones** you aim to complete.
- Your **initial approach** to tackling the problem.

5. Data & Resources Describe the dataset(s), APIs, models, or tools you plan to use. Not all projects require a traditional dataset — if your project centers on prompt engineering, agent design, or model comparison, describe the resources and benchmarks you will rely on instead. For publicly available datasets, consider:
🔗 https://datasetsearch.research.google.com/

Deliverables:

- Each team needs to submit a PDF file containing your Final Project Proposal *(one submission per team)*
- Signed Team Contract (by all members) using the provided template *(one submission per team)*

M2: Progress Presentation

By this milestone, you've been working on the project for 3–4 weeks. Each team will prepare an asynchronous progress presentation consisting of a **short video with slides** and a **poster**.

Video (3–4 minutes max):

- **Project overview:** Brief recap of your research question and why it matters.
- **Progress so far:** What you've built, tested, or discovered. Show preliminary results, demos, or visualizations where possible.
- **[Optional] Challenges & pivots:** What obstacles have you encountered? Have you changed direction from your original proposal?
- **Plan for the remaining time:** What specific tasks remain and how will you divide them?

Keep it focused — prioritize showing your work (results, demos, figures) over repeating your proposal.

Poster: Prepare a single-page poster summarizing your project's current state. Think of it as a visual companion to your video — something a classmate could glance at and quickly understand your project and where it stands. I will be providing you with some templates of my own for your convenience.

Async Engagement: We are finalizing a platform for teams to view each other's presentations and provide feedback during Week 13. Details will be announced on Canvas/Piazza.

Evaluation Criteria: Your M2 will be evaluated on clarity of communication, technical depth of progress shown, effective use of visual aids, organization, time management, and poster quality.

Deliverables:

- Each Video demonstration with slides (3–4 minutes max, one per team) — upload link or file as instructed on Canvas *(one submission per team)*
- Poster as a PDF *(one submission per team)*

M3: Final Project Report & Code

Your final deliverable is a detailed research report and code submission presenting the full scope of your Generative AI project. As an Applied Research project, your report should demonstrate both a working system and rigorous analysis — not just what you built, but what you learned.
The report should follow a structured, professional format using the sections below.

1. Introduction:

- Present the context and significance of your project,
- Clearly define the problem and its relevance to Generative AI, and state your research question and objectives.
- This section should set up the entire paper — a reader should understand *what* you did and *why* after reading it.

2. Literature Review:

- Review existing research related to your problem.
- Summarize key prior work, methodologies, and findings, and identify the gaps or limitations your project aims to address.
- Minimum 6-7 relevant references from peer-reviewed or reputable sources.

3. Methodology:

- Describe the approaches and techniques you used,
- Justify your design choices and discuss any modifications or pivots from your original proposal.
- Include enough technical detail for reproducibility.
- If you explored multiple approaches (which you should), explain each and why you chose to compare them.

4. Experimental Results & Discussion:

- Present your findings with supporting data, tables, and figures.
- Interpret results in the context of your research question.
- Equally important: discuss challenges, limitations, unexpected observations,
- What you would do differently with more time.

5. Conclusion & Future Work:

- Summarize your key findings and their impact (one paragraph).
- Propose concrete next steps for extending the work (one paragraph).

6. References: Provide a properly formatted list of all cited sources. Follow a consistent citation style (NeurIPS format preferred, APA or IEEE also acceptable).

Format and Presentation:

- For a consistent, professional, and organized structure, we'll use the well-known NeurIPS 2024 (or 2025) latex template:
    1. Sign up for Overlead (it's free)
    2. Open the template: https://www.overleaf.com/latex/templates/neurips-2024/tpsbbrdqcmsh
    3. Start creating…

Deliverables:

📌 **Final Report (PDF) -** One submission per team

- Length: **8-12 pages** (6 pages would be too short, 14 pages would be too long), following the NeurIPS template.
- Must include **tables** and **figures** to summarize ideas effectively and support findings.

📌 **Code Submission -** One submission per team

- Preferably, provide a **GitHub repository** or **Google Colab (Jupyter) Notebook link**.
- Alternatively, submit a **zip file** containing your code.
- **README File** and **requirements.txt**:
    - Overview of your code structure.
    - List of any **libraries/packages** used and their versions.

📌 **Peer/Self-Assessment -** Individual Submission

- Each team member must **complete a peer assessment** to evaluate contributions.